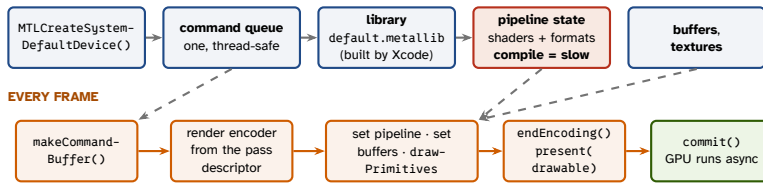


**In one line:** Metal is Apple's low-overhead GPU API. You build the **expensive** objects **once** — device, command queue, **pipeline states** compiled from MSL shaders, buffers, textures — and every frame **record** commands into a transient **command buffer** through **encoders** (render / compute / blit), `commit()` it, and the GPU runs it **asynchronously** while the CPU prepares the next frame. Correctness then hinges on one rule: the CPU must not overwrite data the GPU is still reading.

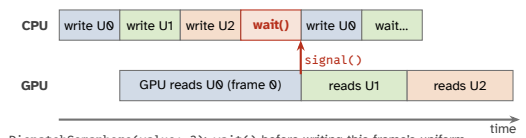
#### A. Build once → record every frame

##### ONCE (load)



Encoders: **render** (draws into attachments) · **compute** (dispatchThreads) · **blit** (copies, mipmaps). One open at a time; a command buffer is single-use. `addCompletedHandler` fires on a background thread when the GPU is done.

#### B. Three frames in flight, one semaphore



DispatchSemaphore(value: 3); `wait()` before writing this frame's uniform buffer `U[frame % 3]`; `signal()` in `addCompletedHandler`. The CPU can run at most 3 frames ahead and never touches a buffer the GPU is reading — without it: flicker and torn data. Every early return must signal too, or the app hangs.

### How it works

- Lifetimes:** device, queue, library, pipeline states (render / compute) and big resources are long-lived; pipeline compilation is the classic first-use hitch — build at load (or cache with `MTLBinaryArchive`). Command buffers and encoders are cheap and single-use.
- Storage modes:** `.shared` — CPU + GPU, one copy in unified memory; `.private` — GPU only, fastest for static meshes and render targets (fill via a blit); `.memoryLess` — lives only in tile memory during a pass (depth, MSAA): no RAM at all. Single-use data under **4 KB**: `setVertexBytes`, no buffer.
- Tile-based GPUs:** Apple GPUs shade on-chip tiles, then write out. A pass's `loadAction` (`.clear` / `.dontCare` vs `.load`) and `storeAction` (`.dontCare` vs `.store`) decide the memory traffic — never load or store what you do not need.
- MSL** is C++-based: vertex, fragment and kernel functions; bindings by attribute — `[[buffer(n)]]`, `[[texture(n)]]`, `[[vertex_id]]`, `[[stage_in]]`, `[[position]]`, `[[thread_position_in_grid]]`. Clip space: `x, y` in `-1...1` (`y` up), depth `0...1`; texture coordinates start top-left.
- MTKView** (MetalKit) wraps a `CAMetalLayer`. Modes: timer (default, `preferredFramesPerSecond = 60`) · on `setNeedsDisplay` (`isPaused + enableSetNeedsDisplay`) · explicit `draw()`. Reading `currentRenderPassDescriptor` acquires the drawable — do it **as late as possible**; the drawable pool is tiny.
- Background:** iOS refuses GPU work from a backgrounded app (the command buffer fails) — pause on `resign-active`. iOS 26 exception: `BGContinuedProcessingTask` + the `background-GPU` entitlement, where `supportedResources` includes `.gpu`.
- Metal 4** (iOS 26, A14 / M1 and later), opt-in beside the classic API: `MTL4CommandQueue`; command buffers made by the device with your own `MTL4CommandAllocator`; `MTL4ArgumentTable` for bindings; `MTL4Compiler` for explicit shader compilation; residency sets; `MTLTensor` + an ML encoder; MetalFX frame interpolation.

### Example — the frame loop (MTKViewDelegate)

```
func draw(in view: MTKView) {
    inFlight.wait() // at most 3 frames ahead
    let u = uniforms[frame % 3]; frame += 1 // THIS frame's copy only
    write(time: CACurrentMediaTime(), into: u)
    guard let cb = queue.makeCommandBuffer()
    let rpd = view.currentRenderPassDescriptor, // drawable: late
    let enc = cb.makeRenderCommandEncoder(descriptor: rpd),
    let drawable = view.currentDrawable
    else { inFlight.signal(); return } // never leak a slot
    enc.setRenderPipelineState(pipeline) // built once, at load
    enc.setVertexBuffer(u, offset: 0, index: 0)
    enc.drawPrimitives(type: .triangle, vertexStart: 0, vertexCount: 3)
    enc.endEncoding(); cb.present(drawable)
    cb.addCompletedHandler { [inFlight] in inFlight.signal() }
    cb.commit() // GPU starts; CPU moves on
}
```

### Shaders — vertex, fragment, compute (MSL)

```
struct VOut { float4 pos [[position]]; float4 color; };
vertex VOut vmain(uint id [[vertex_id]],
                  constant float2 *pts [[buffer(0)]] {
    return { float4(pts[id], 0, 1), float4(1, 0.5, 0, 1) }; }
fragment float4 fmain(VOut in [[stage_in]]) { return in.color; }
kernel void invert(texture2d<float, access::read> src [[texture(0)]],
                  texture2d<float, access::write> dst [[texture(1)]],
                  uint2 gid [[thread_position_in_grid]]) {
    if (gid.x >= dst.get_width() || gid.y >= dst.get_height()) return;
    float4 c = src.read(gid); dst.write(float4(1 - c.rgb, c.a), gid); }
```

### SwiftUI shaders (iOS 17) — Metal without the setup

`[[ stitchable ]]` MSL functions from `ShaderLibrary`; SwiftUI passes position (+ colour or layer), you pass `.float`, `.float2`, `.color`, `.image` arguments:

```
.colorEffect          half4 f(float2 position, half4 color, args...)
.distortionEffect      float2 f(float2 position, args...); source point
.layerEffect           half4 f(float2 position, SwiftUI::Layer l, args...)

#include <metal_stdlib>
#include <SwiftUI/SwiftUI_Metal.h>
using namespace metal;
[[ stitchable ]] half4 wave(float2 p, SwiftUI::Layer layer, float t) {
    return layer.sample(float2(p.x, p.y + 6 * sin(p.x / 20 + t))); }
Text("Hello").layerEffect(ShaderLibrary.wave(.float(t)), // Swift
                           maxSampleOffsetSet: CGSize(width: 0, height: 6))
```

### Where an app developer meets Metal

SwiftUI shaders · Core Image kernels · camera / video frames as textures (`CVMetalTextureCache`) · MPS, `MPSGraph`, Core ML on the GPU · `SpriteKit`, `SceneKit`, `RealityKit`, `drawingGroup()` · games (MetalFX).

### Traps

- ! Pipeline or queue built per frame — compile stalls.
- ! CPU rewrites a buffer the GPU still reads — flicker.
- ! Early return without `signal()` — frozen after 3 frames.
- ! `waitUntilCompleted()` on main — CPU, GPU in series.
- ! `.load` / `.store` on every attachment — wasted bandwidth.
- ! Sampling off position with no `maxSampleOffset`.

### Remember

**“Compile once, record every frame, commit and let go — three buffers, one semaphore.”**